

Vasile Grafu

Engineering Manager | Solutions Architect | Enterprise & AI Systems

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SUMMARY

Engineering leader with 20+ years in software — from real-time game programming to cloud platforms handling millions of requests a day, now building agentic AI systems. I lead one of nShift's core engineering teams, design the architecture behind it, and still write the code myself.

- Millions of requests served every day by the platform my team runs
- 50 web services in production, distributed across AWS
- 10,000+ business clients on the systems I have led
- 20+ years, from real-time game engines to agentic AI systems

SKILLS

Leadership & Management: Engineering Management, Team & Technical Leadership, Hiring & Mentoring, Agile / Scrum, Project Management (Prince2), Strategic Planning

Backend & Architecture: C#, .NET, ASP.NET Core, Entity Framework, LINQ, Python, FastAPI, SQLAlchemy, Solution & Software Architecture, Framework Design, API Design (REST, gRPC, SOAP/WCF)

Frontend & Web: JavaScript, TypeScript, ReactJS, Material UI, HTML, CSS, SASS, Astro, Tailwind CSS, HTMX, Server-Rendered Web (ASP.NET MVC, FastAPI)

AI & Machine Learning: AI-Assisted Software Development (Claude Code), Advanced Agent Skills & Prompt Engineering, Agentic Architectures & LLM Orchestration, RAG (Retrieval-Augmented Generation), MCP Gateways, Servers & Integrations, Machine Learning, Neural Networks & Deep Learning Foundations

Cloud, Data & DevOps: AWS (EC2, ECS, RDS, S3, CloudFormation), Docker, CI/CD, SQL Server, PostgreSQL, Database Design, Relational Databases

Foundations: Probability, Linear Algebra, Calculus, Statistics, Data Structures & Algorithms, Problem Solving & Analytical Thinking, Software Architecture Principles, Design Patterns & Clean Code

EXPERIENCE

Engineering Manager · nShift, Bucharest, Romania

January 2022 – Present

Global delivery-management platform — over 1,000 carriers across some 190 countries.

I lead the Webservices Team — one of the company's core teams — building web services that handle millions of requests per day and support multiple business-critical applications. I combine hands-on development with leadership: contributing to coding, architecture, and decision-making while ensuring alignment across teams.

- **Lead and manage the Webservices Team:** setting clear goals, providing direction, and keeping every member motivated and effective.
- **Establish engineering processes and standards:** consistent development practices that improve collaboration, maintain quality, and accelerate delivery.
- **Recruit and develop engineers:** active role in hiring, onboarding, and mentoring to build a skilled, cohesive team.
- **Provide technical guidance:** solving complex issues alongside developers, reviewing designs, and making critical decisions on challenging topics.
- **Contribute to software architecture:** design and implementation of scalable, secure, and maintainable systems.
- **Hands-on development:** shared components, application structure, data flow, database design, security principles, and infrastructure improvements.
- **Solution Architect for our applications:** designing and managing cloud environments on AWS and implementing CI/CD pipelines.

- **Act as Scrum Master:** stand-ups, planning, retrospectives, and effective communication between Product Owners and developers.
- **Building a large-scale conversational AI platform:** LLMs, agent orchestration, RAG, and MCP-style tool integration.

AI-assisted development (Claude Code) · LLMs, Agents, RAG, MCP · C# / .NET / ASP.NET Core · Python / FastAPI / SQLAlchemy · SQL Server / PostgreSQL · AWS (EC2, ECS, RDS, S3) · Docker, CI/CD · JavaScript / TypeScript / ReactJS / Material UI

Team Manager · Consignor, Bucharest, Romania

January 2017 – January 2022

Scandinavia's leading multi-carrier shipping software; 10,000+ business customers.

I led two teams — the Webservices Team and the Integration Team — and wore the hat each situation demanded: Team Manager, Technical Lead, Project Manager (Prince2), or Software Developer. In practice that meant hiring and training the people who joined, owning the critical architecture decisions, running projects across team boundaries — and still writing code myself.

- **Webservices Team:** ran the web services serving the client components of the Consignor business.
- **Integration Team:** brought Consignor functionality to the major e-commerce platforms.
- **Team leadership:** set goals and direction for both teams, keeping two groups with very different missions pulling toward one platform.
- **Technical guidance and architecture:** made the critical design decisions and worked through the hard problems alongside the developers.
- **Recruit and develop engineers:** hiring, onboarding, and training the people who joined either team.
- **Hands-on development:** shared components, application structure, and database design across the Consignor suite.

C# / .NET web services · E-commerce platforms (WooCommerce, Klarna, Magento) · APIs & integrations · AWS · Project management (Prince2)

Technical Lead · Consignor, Bucharest, Romania

2009 – 2017

I led the Webservices Team, behind the 50 web services that served over 10,000 business clients in the Consignor application suite, distributed across Amazon Cloud. My role was to find sound technical solutions, mentor and train developers, help hire the people who joined us, and — not least — write a great deal of the code myself.

- **End-to-end delivery:** involved across the entire application development workflow.
- **Module design:** designed many modules of the business software to satisfy client requirements.
- **Mentoring:** guided both junior and senior developers.
- **Documentation:** wrote detailed technical and user documentation.
- **Cross-department liaison:** connected the development team with other departments.

C# / .NET, ASP.NET MVC · WCF web services · SQL Server Enterprise · JavaScript, jQuery, Bootstrap, LESS/SASS · Python · AWS

Team Leader · TeamNet, Bucharest, Romania

2008 – 2009

Romanian technology and systems-integration company.

- **Optimus ERP modules:** managed them through every development step, from specification to testing and deployment.
- **Feature planning:** built development plans for new features with the design department.
- **Quality and maintenance:** enhanced existing features, and triaged and resolved every defect raised by the Quality Department.

C# / .NET · Windows Forms · SQL Server

Game Programmer · Ubisoft, Bucharest, Romania

January 2005 – June 2008

One of the world's largest video game publishers.

I designed, developed, and optimized gameplay and platform modules on four shipped and in-development titles.

- **Silent Hunter 4:** designed and built the framework managing the evolution of the Crew, Weapons, and Equipment of the U-Boat simulator, plus the Career module.
- **King Kong (PC):** Windows Media Center integration, multithreaded rendering, and the security module.
- **Blazing Angels (PC):** Save/Load system and Media Center shell implementation.
- **Tom Clancy's FireHawk:** ERS (Enhanced Reality System) module.

C++, STL · Gameplay & platform modules · Multithreaded rendering · Software optimization · Linear algebra, calculus, probabilities

KEY PROJECT

Conversational AI Platform

Engineering Manager & Architect · nShift

A large-scale AI assistant built on an agentic architecture: LLMs combined with orchestration layers, RAG, and structured prompt engineering. Domain-specific agents for different departments — each with tailored prompts, tools, and data access — collaborate through an orchestration layer. A continuous ingestion pipeline indexes internal documentation, APIs, and business data into knowledge stores, so responses stay context-aware and grounded in company-specific information.

EDUCATION

Master of Science in Computer Science, Specialization in Project Management — Academy of Economic Studies, Bucharest (2000 – 2002)

Bachelor in Business Administration, Major in Computer Science, Specialization in Computer Science Applied in Business — Academy of Economic Studies, Bucharest (1995 – 2000)

SELF-LEARNING

- Studied machine learning, deep learning, neural networks, probability, linear algebra, calculus and statistics independently — to understand the mathematics, not just the frameworks.
- Prefer to understand mechanisms, assumptions, limitations and trade-offs rather than treat algorithms and models as black boxes.
- That habit is what lets me reason about system design and model behaviour in areas I have not worked in before.